19 · Case 14, part 2: Simon's family

The words of this track, as it says them. Bold brown is the examiner.

Exam case 14, continued. Simon's family.

Seven marks. Draw a pedigree chart showing Simon, his mum, his dad, his maternal grandmother, who is not colour blind, his maternal grandfather, who is colour blind, his maternal aunt, who is colour blind, and his maternal uncle, who is not. Can you track how Simon inherited it, when neither of his parents is colour blind?

Write each genotype on the X chromosome. X big B is normal vision. X little b carries the colour blindness. A man has one X and a Y. A woman has two X's.

The hook: Grandad's X little b goes to a carrier mum, and then to Simon. His dad only gives him a Y.

His maternal grandfather: X little b, Y. A man, colour blind, and the condition is recessive.

His maternal grandmother: X big B, X little b. A carrier. She isn't colour blind, but she has a colour-blind daughter, so she must carry one.

His maternal aunt: X little b, X little b. A colour-blind woman has the defect on both of her X chromosomes.

Simon's mum: X big B, X little b. A carrier. She passed it to Simon, but her normal X big B gives her normal vision.

Simon's dad: X big B, Y. He isn't colour blind, so he has the normal X.

His maternal uncle: X big B, Y. Not colour blind.

And Simon: X little b, Y. His X little b came from his mum. His dad gave him only a Y.

Six marks. Given all that, how is red-green colour blindness inherited?

X-linked recessive. The genes for the L and M cones, red and green, are on the X chromosome. It's often carried by mothers, and expressed in fifty per cent of their sons. It only shows when all the X chromosomes are affected, so a woman needs both. Fathers never pass it to their sons, because they only give their sons a Y. Fathers always pass the affected X to all of their daughters. And it can skip a generation.

That's her "better answer", from her post on what to study for the exam.

Two marks. Will Simon pass red-green colour blindness on to his children? How?

To none of his sons: they get his Y. And to all of his daughters, as unaffected carriers: each daughter gets his X little b. Her quiz key, word for word: to none of his sons, but to all his daughters, who will be unaffected carriers.

And her quiz asks for his mum's genotype as a multiple choice. The answer: X little b, X big B. She carried it over to Simon, but her normal X big B gave her normal vision.

Once more. Simon's family.

Grandfather, X little b, Y: colour blind. Grandmother, X big B, X little b: a carrier, because she has a colour-blind daughter. Aunt, X little b, X little b. Uncle, X big B, Y. Mum, X big B, X little b: a carrier, with normal vision. Dad, X big B, Y. Simon, X little b, Y: his X came from his mum, and his Y from his dad. It's X-linked recessive: carried by mothers, shown in half of their sons, a woman needs both X's, fathers never pass it to sons and always pass it to daughters, and it can skip a generation. Simon will pass it to none of his sons, and to all of his daughters, as unaffected carriers.